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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

8644695785

ADDITIONAL MATHEMATICS **4037/11**

Paper 1 **May/June 2011**

2 hours

Candidates answer on the Question Paper.

No additional materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.
The total number of marks for this paper is 80.

For Examiner's Use	
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Total	

This document consists of **16** printed pages.

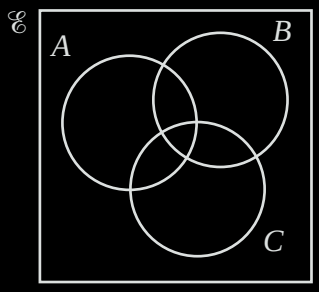
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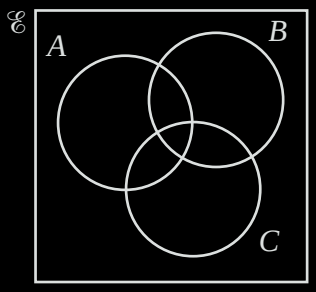
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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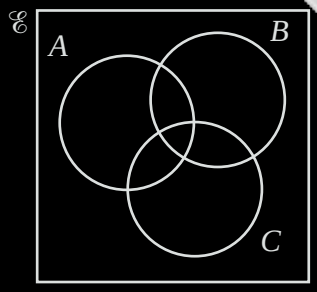
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$(A \cup B) \cap C'$



$(A \cup B \cup C)'$



$(A \cap B) \cup (B \cap C) \cup (C \cap A)$

[3]

(b) Given that $P = \{p : \tan p = 1 \text{ for } 0^\circ \leqslant p \leqslant 540^\circ\}$, find n($P$). [1]

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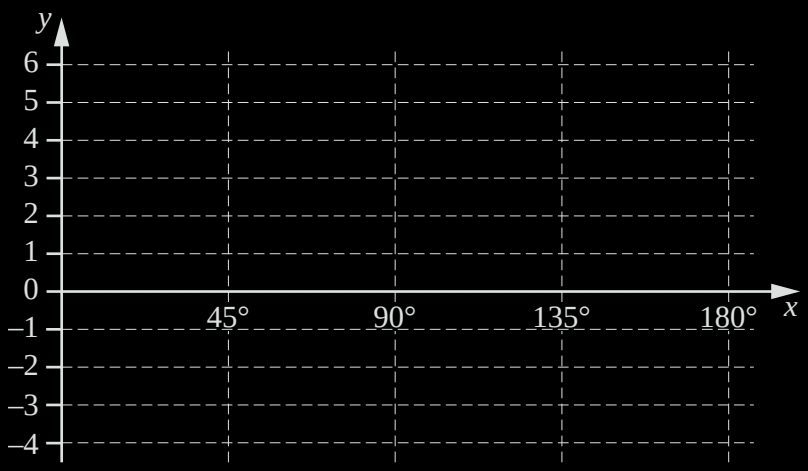
4 **(a)** Solve the equation $16^{3x-2} = 8^{2x}$. [3]

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$\dfrac{a^{\frac{4}{3}}b^{-\frac{2}{5}}}{a^{-\frac{1}{3}}b^{\frac{3}{5}}} = a^p b^q$, find the value of p and of q. [2]

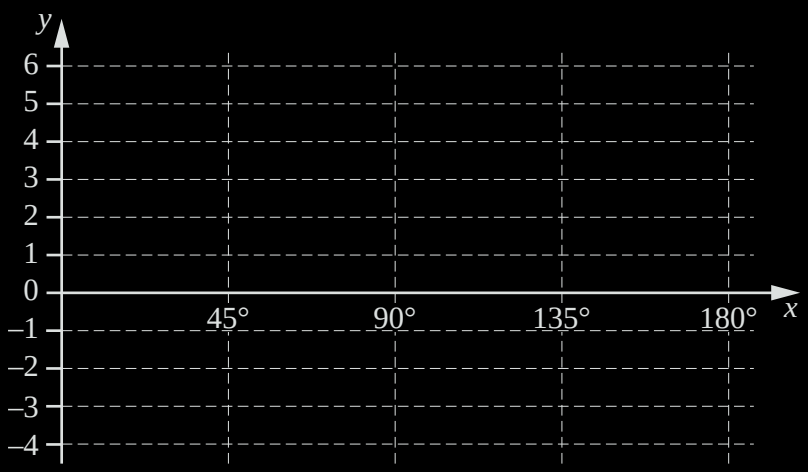
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5 **(i)**



On the diagram above, sketch the curve $y = 1 + 3\sin 2x$ for $0^o \leqslant x \leqslant 180^o$. [3]

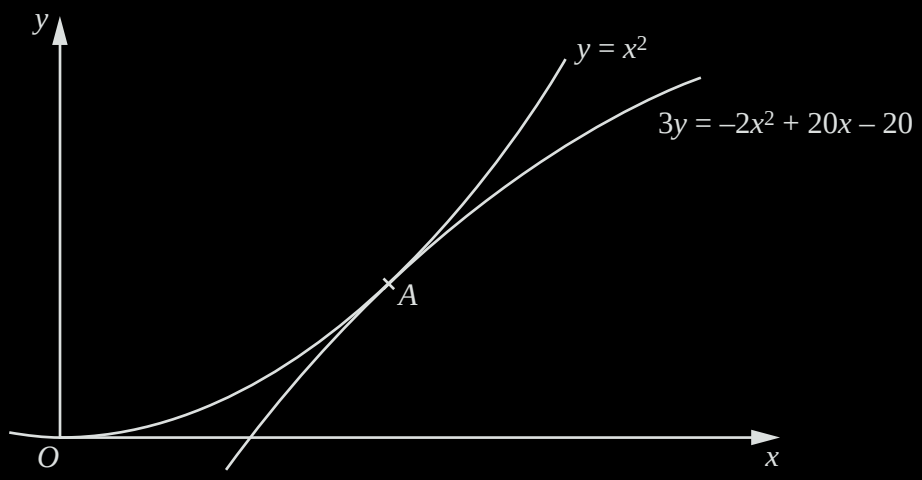
(ii)



On the diagram above, sketch the curve $y = |1 + 3\sin 2x|$ for $0^o \leqslant x \leqslant 180^o$. [1]

(iii) Write down the number of solutions of the equation $|1 + 3\sin 2x| = 1$ for $0^o \leqslant x \leqslant 180^o$. [1]

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(i) Show that the x-coordinate of A is 2. [1]

(ii) Show that the gradients of the two curves are equal at A. [3]

(iii) Find the equation of the tangent to the curves at A. [1]

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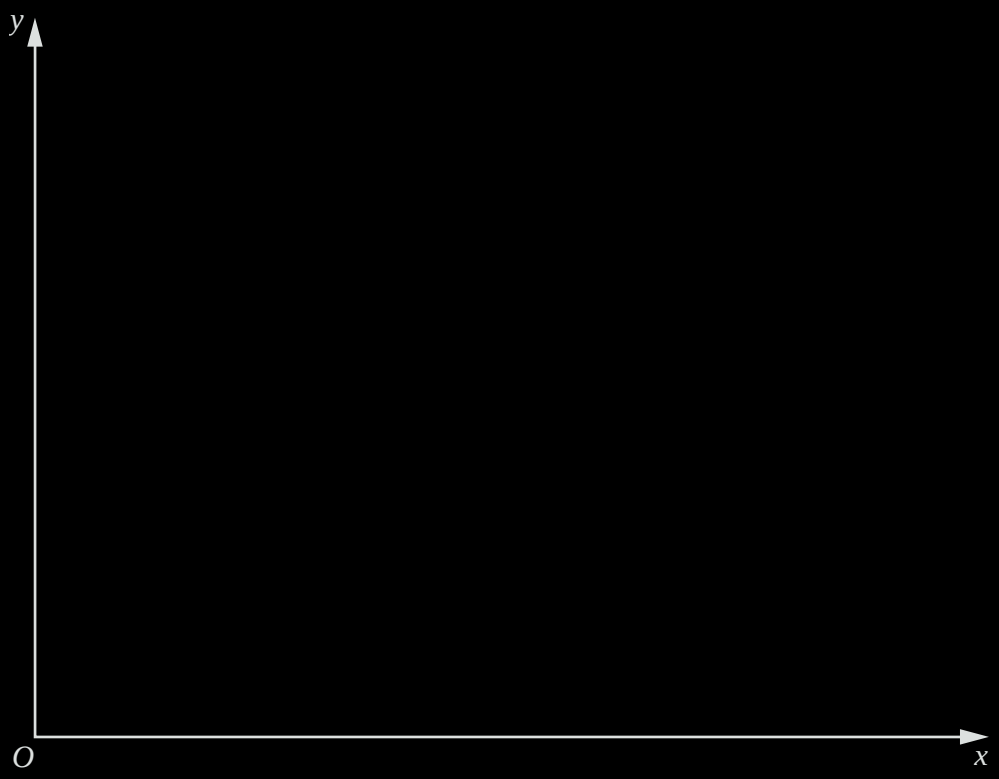
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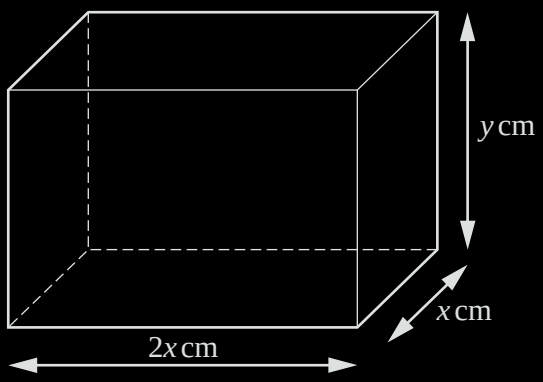
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(iii) Find the value of x for which $\mathrm{gh}(x) = 85$. [4]

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(i) Show that the surface area $A\,\text{cm}^2$ of the cuboid is given by

$$A = 4x^2 + \frac{216}{x}.$$

[3]

(ii) Given that x can vary, find the dimensions of the cuboid when A is a minimum. [4]

(iii) Given that x increases from 2 to $2 + p$, where p is small, find, in terms of p, the corresponding approximate change in A, stating whether this change is an increase or a decrease. [3]



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